Utilizing markerless motion capture for repositioning human body models

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Introduction: The Global Human Body Models Consortium (GHBMC) are Finite Element (FE) computational models of the human body used largely in a research capacity to simulate the biomechanics and kinematics of the human body in a vehicle crash. There are a number of models ranging from male and female, from 05th, 50th, and 95th percentile. These GHBMC models are used in validation test simulations, in order to validate an experimental impact test, the GHBMC model needs to be positioned similarly to the subject or Anthropomorphic Test Devices (ATD) in the test. Therefore, there is a lot of crucial information needed such as joint angles and distance from impactor to get accurate simulation results. This study elaborates on developing a semi stereo vision, markerless motion capture method using machine learning techniques for human pose estimation to reposition GHBMC models.

Materials and Methods: Python and the YOLOv7 machine learning detection algorithm [1] was primarily used to develop the repositioning tool. YOLOv7 was used for the human pose estimation[2], integrating with a semi stereo vision. Semi stereo view was created by setting up two cameras perpendicular to each other to get an and y-axis in one camera and a z and y-axis in the other. However, this is not stereo view is because the cameras are not triangulating the points into real space during real time. Therefore, a semi stereo view was chosen as a proof of concept. A subject stood in frame with their full body in view and their nose, eyes, ears, shoulders, elbows, wrists, hips, knees, and ankles. In order to save out the coordinates the YOLOv7's run_pose.py was modified to output all the key points coordinates along with its confidence interval into a .csv file. A python script was then created to read the X, Y and Z coordinated from camera 1 and camera 2 at each frame and transform the acquired 2D coordinates into 3D. Joint angles were then calculated from the skeletal motion which is then applied to the primitive pedestrian model for repositioning which is, an essential step for improving finite element simulations.

Results and Discussion: The implementation of YOLOv7 markerless motion capture and the semi stereo vision yielded a positive outcome in using human pose estimation for GHBMC repositioning. By using machine learning techniques and a two camera perpendicular set up, the system is able to convert 2D coordinates into useful 3D data such as key point positions and joint angles. This method was able to successfully reposition the primitive pedestrian model and can therefore, be used in other GHBMC models.

Conclusion: The results from this study suggest a promising foundation that markerless motion capture can successfully be used as a proof of concept to reposition GHBMC model for FE simulations. This underlines the possibility of using machine learning for subject specific skeletal motion tracking. The use of a subject-specific approach significantly increases the authenticity and applicability of simulation data, which can be critical in various domains, from medical procedures to athletic performance evaluation. A more advance full stereo vision set up is also possible for better depth perception and more accurate coordinates which leads to improved pose estimation and ultimately improved model repositioning. A challenge still remains in properly implementing in real world applications because there are a number of variable to consider that will affect the data such as lighting conditions, subject's article of clothing, and other environmental factors. The research's successful conclusion prompts the idea of further investigation of markerless motion capture for model repositioning.

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